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OPTICAL GLASS

This is an optical glass brevier for quick reference. Based on our comprehensive catalog 3050e and its supplements up to June 1968, it provides compact information on the basic properties of all our optical glass types.

When additional design data are required, our main catalog with more than 25.000 numerical values will answer all pertaining questions.

SCHOTT OPTICAL GLASS Inc., DURYEA, PENNSYLVANIA 18642

1. Designation of the Optical Glass Types

An optical glass type is designated according to its position in the n_d/ν_d diagram. The Abbe-number ν_d is defined as $\nu_d = \frac{n_d - 1}{n_F - n_C}$. The index difference $n_F - n_C$ is called the main dispersion, but when $n_{F'} - n_{C'}$ is used as the main dispersion, the Abbe-number reads $\nu_e = \frac{n_e - 1}{n_{F'} - n_{C'}}$.

In the n_d/ν_d diagram the glasses are divided into groups. The reference to each glass type of the diagram is composed of the abbreviated group designation and a number. In addition to our traditional glass type designations, this catalog lists 6 numbers the first 3 of which indicate the refractive index n_d and the second 3 the Abbe-number ν_d (Example: BK 7 $n_d = 1.51680$ and $\nu_d = 64.17$ is shown as BK 7 – 517642).
The glass groups are listed according to decreasing ν_d-values.

Traditionally the glass types with $n_d > 1.60$, $\nu_d > 50$ and $n_d < 1.60$, $\nu_d > 55$ are designated as "Crowns" (K) and the remaining types as "Flints". The "K" grouping in the diagram also includes the barium light crowns (BaLK) and the zinc crowns (ZK). Special signs further identify short flints (KzF) and special short flints (KzFS). The optical values of some LaK and LaSF types could be achieved only by addition of thorium oxyde. These

few glass types are denoted by "Th" in the column "Special Characteristics", and the Th-content in % is given in brackets.

2. Optical Properties

a) Refractive Index and Dispersion

For the following 9 spectral lines, refractive indices with 6 decimals are listed or can be derived from the partial dispersions which are also mentioned.
(Our main catalog No. 3050e gives information on 13 spectral lines)

Spectral Line	violet mercury line	blue mercury line	blue cadmium line	blue hydrogen line	green mercury line	yellow helium line	red cadmium line	red hydrogen line	red helium line
	h	g	F'	F	e	d	C'	C	r
Element	Hg	Hg	Cd	H	Hg	He	Cd	H	He
Wave-length in nm	404.66	435.84	479.99	486.13	546.07	587.56	643.85	656.27	706.52

FK PK
PSK BK
BaLK
K
ZK BaK
SK
KF BaLF
SSK
LaK
cont'd

The refractive indices listed for each glass type represent the mean values of several melts.

To eliminate the possibility of systematic errors, all measurements were carried out by two different methods. The dispersion ($n_{\lambda_1} - n_{\lambda_2}$) could be determined to $\pm 3 \times 10^{-6}$.

Melt data sheets are sent with each delivery. Our **"standard measurement"** provides an accuracy of $\pm 3 \times 10^{-5}$ for the refractive indices and of $\pm 2 \times 10^{-5}$ for the dispersions. Refractive indices and dispersion data are given to 5 decimal places. For **"precision measurements"** the accuracy is $\pm 1 \times 10^{-5}$ and $\pm 3 \times 10^{-6}$ respectively. In this case the refractive indices are given to 5 decimal places and the dispersion values to 6 decimal places. Temperature and atmospheric pressure are stated in the data sheets.

An accuracy of $\pm 5 \times 10^{-6}$ for the refractive indices and of $\pm 2 \times 10^{-6}$ for the dispersions can be provided at increased expenditure.

3. Chemical Properties

a) Resistance to Climatic Variations

Due to the influence of the water vapor in the air and especially under high relative humidity and higher temperature, a change in the surface of the glass may occur and usually appear as a cloudy film which cannot be wiped off. Under normal atmospheric conditions, even on sensitive glasses, these changes occur mostly at a very slow rate.

Therefore, to test glasses for stability under climatic variations, an accelerated procedure is followed, whereby polished, uncoated glass plates are exposed to a vapor-saturated atmosphere the temperature of which changes at an approximate hourly rhythm between 40 °C and 50 °C. In this process, a condensation on the surface of the glasses occurs during the heating phase. This disappears during the cooling phase. After exposure times of 30, 100 and 180 hours, the glasses are removed from the climatic chamber. The classification in four groups is then obtained by diffused light measurements of the treated surfaces.

Glasses of group 1, even after a seven day exposure under this test for climatic variations, show no attack or only a negligible one. Under normal humidity conditions during working and storage, a surface attack on group 1 glasses is not to be expected. On glasses of group 4, however, a marked attack occurs a few hours after exposure under the conditions described. Therefore, great care should be taken during processing, storage and assembly. Groups 2 and 3 denote, in numerical order, grades of sensitivity between these two limits.

b) Resistivity to Staining

The classification of resistivity to staining is determined by the following method: The flat polished glass sample is pressed onto a cuvette with a spherical hollow of a maximum depth of 0.25 mm, which contains a few drops of the test solution.

Test solution I: Standard acetate $p_H = 4.6$
Test solution II: Sodium acetate buffer $p_H = 5.6$

Through the action of the test solution a decomposition takes place on the glass surface, and stains of interference colors are formed more or less rapidly. The glasses are classified according to the time after which the brown-blue color alteration appears under a temperature of 25 °C.

Group 0 of resistivity to staining indicates all glasses which show practically no susceptibility to staining even after a 100-hour-long exposure to test solution I.

All glasses showing the color formation before 100 hours of exposure, belong to groups 1 – 5, group 1 being the slowest and group 5 the quickest to react to staining.

On group 5 glasses, the color formation appears in less than 12 minutes by exposure to both test solutions I and II. These glasses must, therefore, be handled with great care during processing.

The classification of resistivity to staining, however, does not give any indication as to the resistance to climatic variations.

c) Acid resistance

As desired by many customers, we mention, in addition to "Resistivity to Staining", the classification "f" of "Acid Resistance" in the 5th column under "Chemical Stability". The following definition is applicable:

Due to their basic components (alkaline oxides, alkaline earth oxides, lead oxide, etc.) glasses on the whole are attacked by acids, but this attack remains insignificantly small for the majority of optical glasses during processing and use. If the glass contains insoluble components, especially silicic acid in sufficient quantity, these remain on the surface after removal of the soluble components in the form of insoluble, protective layers. These layers are exceptionally cohesive and completely optically homogeneous, tending to prevent any further attack. If the thickness of these protective layers, however, reaches 1/4 wave-length, coloured spots appear due to interference, a phenomenon which in the optical industry is often called oxidation, since the colours have a superficial similarity to oxide layers on metals. With extremely sensitive glasses, oxidation takes place without any residue of a protective layer. A measure of the acid attack is given in all cases by the thickness of the destroyed glass film. No account is taken here of the strong luminosity of the coloured spots on high refractive glasses, which are therefore more noticeable than on low refractive glasses.

The acid resistance of the glasses against strong acids is determined by the attack of 0.5 n HNO_3 ($p_H = 0.3$) at 25 °C. The time t in hours which is required for the destruction of a glass film of 0.1 μ thickness gives the indication of the 5 classes of acid resistance.

Class	f 1	f 2	f 3	f 4	f 5
Time t in hours for 0.1 μ ($p_H = 0.3$)	> 100	100—10	10—1	1—0.1	< 0.1

Unfortunately with many glasses and particularly with just those of special interest to the optical designer, it is impossible to add insoluble components in such quantities as necessary for the formation of a protective layer of sufficient thickness without having an unfavourable effect on the optical values. Such glasses are therefore very sensitive

during processing, since they can be attacked even by very weak acids, p_H 4—7 (e. g. perspiration, cements, CO_2 etc.). All of them belong to class 5, and are, in addition, designated by their behaviour to n-acetate ($p_H = 4.6$).

The time t required to destroy a layer of thickness of 0.1 μ at a temperature of 25 °C serves as a measure. Accordingly, the glasses of class 5 are divided into 3 sub-groups 5a, 5b and 5c, as follows:

Class	5 a	5 b	5 c
Time t in hours for 0.1 μ ($p_H = 4.6$)	10—1	1—0.1	< 0.1

d) Weather Resistance

Further information on the chemical stability of optical glasses is given in columns 2 and 3. Besides the classification of "Resistance to Climatic Variations" we list, as far as available, data of **the hydrolytic classes "h"** and the **weathering classes "V_k".**

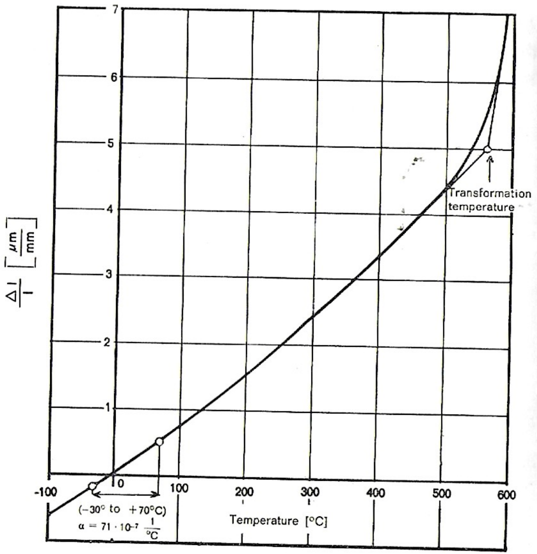
Since the intensity of the weathering effect for silicate glasses is on the whole dependent on their basic components, the method developed by Mylius (Accelerated method) for the **determination of the weathering** alkalinity A_v has proved its value for these glasses, since

4. Thermal Properties

a) Thermal Expansion

The coefficient of thermal expansion of glass is dependent upon temperature. Linear coefficients of expansion are listed which represent mean values for the temperature ranges from −30 °C to +70 °C. The coefficient measured between −30 °C and +70 °C averages at 20 °C, i. e. the temperature at which optical glass is normally used.

Using BK 7 as an example, the following graph shows the effect of temperature on the thermal expansion of glass.



b) Transformation Temperature

The transformation region is that range of temperature in which a glass gradually transforms its solid state into a plastic one. This region of transformation is approximately defined by the transformation temperature Tg (viscosity approximately 10^{13} poise). It is determined from the typical rate of change of thermal expansion in the transformation region, as shown in the graph. The thermal expansion curve is obtained by measuring well annealed glass samples heated at a rate of 4 °C/min.

If heat treatment of optical glass exceeds a temperature of about (Tg − 200) °C, deformations in optical finish and changes of the refractive index may result.

5. Forms of Supply and Quality Definitions

a) Forms of Supply

Slab Glass

is supplied with four sides ground and two opposite surfaces test polished. Edges are slightly bevelled.

Block Glass

is supplied with two opposite surfaces test polished and another two ground; two sides are left fire polished. Edges are rounded.

Strips

are extruded and supplied with fire polished sides.

Rods

have usually a circular cross-section and a fire polished surface.

Gobs

are fire polished blanks of nearly circular cross-section identified by their weight or volume.

Cut Blanks

are forms of glass processed to required sizes and shapes (e. g. plates, discs, prisms).

Pressings

are molded blanks (e. g. lens or prism pressings) made from selected raw glass.

b) Optical Tolerances, i. e. deviations of individual melt data from nominal catalog values

The optical data of individual melts of a glass type vary to a greater or lesser extent from the refractive indices given in this catalog. The tolerance for the refractive index n_d or n_e resp. will not exceed ± 0.001. Only for the very high index glasses SF 57, SF 58, SF 59, LaSF 5, LaSF 6, LaSF 7, LaSF 9, LaSF 12, and LaSF 13 the index tolerance may reach ± 0.002.

The values for v_d or v_e resp. of individual melts may vary by ± 0.8 % from the catalog constants.

Restricted tolerances can be observed for all glass types and forms.

Restricted n_d-tolerances		Restricted ν_d-tolerances	
Grade 1	max. ± 0.0002	Grade 1	max. ± 0.2 %
Grade 2	max. ± 0.0003	Grade 2	max. ± 0.3 %
Grade 3	max. ± 0.0005	Grade 3	max. ± 0.5 %

c) Striae

Striae are undesirable and predominantly thread-like or cord-like vitreous inclusions which, on account of varying chemical composition, also differ in physical properties from the basic glass. Striae are of a very local nature and cause a relatively high index gradient. Our optical glass of "Normal Quality" is examined for striae in one direction by the shadowgraph test. This sensitive shadowgraph method also detects such striae which formerly were hardly or not at all detectable by viewing with the naked eye against a light-dark boundary. The "Normal Quality" may contain isolated fine striae.

The best quality regarding striae, meeting the highest requirements, is provided for glass with "special striae selection" and for "Precision Quality". Striae tests are performed with a highly sensitive shadowgraph method and / or by the Foucault knife-edge test. According to the application, the glass is examined for striae in one or more directions.

Glass types denoted by "p" are particularly prone to fine, parallel striae which, for technical reasons, cannot be avoided. These few glasses should not be used for the manufacture of prisms or thick lenses (over 25 mm approximately). Discs for lenses should be cut from the glass in such a way that the light path is almost perpendicular to the

polished surfaces of the plates supplied. Glass types denoted by "O" **may** contain parallel striae.

The designations "p" or "O" appear in the column "Special Characteristics".

d) Optical Homogeneity (variation of the refractive index)

The degree of refractive index variations within a melt or blank is called optical homogeneity. Excepted from homogeneity specifications are inhomogeneities of a very local nature which were discussed in the preceding chapter "Striae".

A) Glass in Random Size and Cut Blanks

The variation of the n_d-value within one melt will not exceed $\pm 1 \times 10^{-4}$.

With the experience of many years, we developed special production and testing methods which permit us to achieve significantly improved homogeneities.

On special request, we can therefore supply optical glass of the following homogeneity groups:

Homogeneity Group	Maximum Variation of the n_d value	Availability
H 1	$\pm 2 \times 10^{-5}$	within a selected melt
H 2	$\pm 5 \times 10^{-6}$	within a cut blank
H 3	$\pm 2 \times 10^{-6}$	within a cut blank, but dependent on its size
H 4	$\pm 1 \times 10^{-6}$	within a cut blank, but dependent on glass type and size

When inquiring or ordering, please specify exactly the maximum variation of the n_d value permissible for the particular application. Communicate with us in advance on projects requiring large blanks of homogeneity groups H 3 or H 4.

B) Pressings

The variation of the n_d-value from one pressing to another of the same melt and annealing amounts to a maximum of $\pm 2 \times 10^{-4}$. Within the pressing, the variation of the refractive index is markedly smaller.

On special request, we can supply pressings of the following improved homogeneity groups:

Homogeneity Group LH	Maximum Variation of the n^d value within a lens annealing (LK)	Availability
1	$\pm 1 \times 10^{-4}$	within a selected melt
2	$\pm 5 \times 10^{-5}$	On request

e) Stress Birefringence

Optical glass, even after special annealing, retains small residual strains. The amount and distribution of these strains depend on the annealing conditions, dimensions and type of glass. Internal stresses cause a birefringence which may vary according to glass type, the birefringence being measured by optical path differences and specified in nm/cm.

We supply glass blanks with a highly symmetrical stress distribution. The birefringence of cut discs is determined at points located at 5 % from the edge (of the diameter). Rectangular plates are measured 5 % from the edge (of the width) at the middle of the long side.

For optical systems with extremely high performance, a low birefringence is of great importance as well as good optical homogeneity. The development of special annealing methods permits us to achieve very small residual strains, i. e. a very low birefringence, even for glass blanks of very large dimensions.

In the following summary, examples are given of the stress birefringence in cut blanks:

Glass Types and Dimensions	Stress Birefringence		
	Normal Quality	Normal Quality with Special Annealing (NSK) or Precision Quality	Normal Quality with Super-Special-Annealing (NSSK) or Precision Quality with Super-Special-Annealing (PSSK)
All glass types up to sizes of about 150×150×80 mm³	≦ 10 nm/cm		
For all glass types up to ⌀ 300 mm th. 60 mm	≦ 10 nm/cm	≦ 6 nm/cm	≦ 3 – 4 nm/cm according to glass type
For glass types such as BK 7, BaK 4, SK 16, LF 5, F 2, SF 2 up to ⌀ 600 mm th. 80 mm	≦ 12 nm/cm	≦ 6 nm/cm	≦ 4 nm/cm
For glass types such as KzFS 1, KzFS 2 up to ⌀ 400 mm th. 70 mm	≦ 15 nm/cm	≦ 10 nm/cm	≦ 6 nm/cm
For glass types such as BK 7, BaK 4, SK 16, LF 5, F 2, SF 2 up to ⌀ 800 mm th. 100 mm	≦ 15 nm/cm	≦ 10 nm/cm	≦ 6 nm/cm
For glass types such as BK 7, F 2, SF 2, up to ⌀ 1000 mm th. 120 mm	≦ 25 nm/cm	≦ 15 nm/cm	≦ 10 nm/cm

If cut discs or plates are required with even further reduced birefringence values, please contact us. The values mentioned are subject to modifications, if the ratio of thickness to diameter of a blank is larger than outlined.

For extreme mechanical or thermo-mechanical requirements, tempering is advisable to raise the strength of the glass. Cut blanks can be supplied to the highest degree of pre-stressing.

f) Bubbles

Bubbles in optical glass vary in number from one type to another due to the many different compositions and production methods. The bubble content may vary considerably even from melt to melt.

The classification of bubble content is established by specifying, in mm^2, the total of the single cross-sections of bubbles referred to 100 cm^3 of glass. Other inclusions, like small stones or crystals, are regarded as bubbles and included in the count of total bubble cross-section. This classification of bubble content in a glass comprises all bubbles and inclusions of ≥ 0.06 mm.

New production techniques have considerably improved the bubble condition of most optical glasses which can now be classified under a more favorable bubble group. Consequently, a new classification is introduced with 6 groups numbered 0 to 5.

Classification

Bubble Group	Total bubble cross-section per 100 cm^3 volume of glass
0	0 — 0.029
1	0.03 — 0.10
2	0.11 — 0.25
3	0.26 — 0.50
4	0.51 — 1.00
5	1.01 — 2.00

"**Special bubble selection**" can only be ordered for blanks, e. g. cut plates, discs, prisms, and for pressings.

g) Color (Internal Transmittance)

Some optical glasses show a noticeable tint in thicker sections. They absorb light of the various wavelength regions to differing degrees.

By using highly purified raw materials and special melting techniques, the light transmission of our optical glasses was even further improved. However, for certain glasses with extreme optical properties, such as high-refractive dense flints, barium dense flints, lanthanum crowns as well as lanthanum flints and lanthanum dense flints, it is impossible to avoid a yellow or yellow-green tint although raw materials of highest chemical purity are used. Furthermore, the tint of an optical glass melt is not solely dependent on the chemical composition, but also on unavoidable trace impurities. These vary from melt to melt and, even in lowest concentrations of merely 0.00001 %, affect the transmission.

As a guide, details of light transmission are given for each glass type by listing internal transmittance τ_i for the wavelength 400 nm at a thickness of 25 mm. (Our main catalog No. 3050e lists the internal transmittance between 280 to 700 nm)

All internal transmittance data are average values of several melts of a glass type and are meant for general information.

h) Quality Terms

A) Normal Quality (N)

Normal Quality stands for optical glass examined for striae and birefringence in one

direction. The variation of the refractive index n_d does not exceed $\pm 1 \times 10^{-4}$, within one melt.

Normal Quality of Homogeneity Group 1 (NH 1) guarantees a variation of the refractive index n_d not exceeding $\pm 2 \times 10^{-5}$, within one melt.

Normal Quality of Homogeneity Group 2 (NH 2) guarantees a variation of the refractive index n_d not exceeding $\pm 5 \times 10^{-6}$, within one blank.

Normal Quality with Special Annealing (NSK) provides reduction of stress birefringence. If required, the stress can also be raised by special annealing (pre-stressing).

Normal Quality with Special Striae Selection (NVS) is, in striae quality, equivalent to precision quality.

B) Precision Quality (P)

Precision Quality stands for optical glass of extremely careful selection. It is tested, in one or more directions, for striae and birefringence with due regard to its intended application. The variation of the refractive index n_d does not exceed $\pm 5 \times 10^{-6}$, within one blank.

Forms of Supply versus Quality Terms of Optical Glass

Quality (Code)	Forms of Supply					
	Slab Glass	Blocks	Strips	Rods	Gobs	Cut Blanks
Normal Quality (N)	yes	yes	yes	yes	yes	yes
Normal Quality with Homogeneity Group 1 (NH 1)	yes	yes	no	no	no	yes
Normal Quality with Homogeneity Group 2 (NH 2)	no	no	no	no	no	yes
Normal Quality with Special Annealing (NSK)	yes	yes	no	no	no	yes
Normal Quality with Special Striae Selection (NVS)	no	no	no	no	no	yes
Precision Quality (P)	no	no	no	no	no	yes
Special Bubble Selection	no	no	no	no	no	yes
Transmission	Specifications for minimum transmission of a glass type must be submitted for approval, before the order is placed.					

Additional quality requirements other than those stipulated can be met on special request, e. g. cut blanks of precision quality with further improved homogeneity (see page 18).

Note:

The preferred glass types of our production program are marked by heavy print of their six-digit numbers.

Type	n_d	ν_d	n_F-n_C	n_e	ν_e	$n_{F'}-n_{C'}$	n_C-n_r	n_d-n_C	$n_{F'}-n_e$	n_g-n_F	n_h-n_g
FK 1 471 673	1.470690	67.34	0.006990	1.472358	67.14	0.007035	0.001249	0.002157	0.003556	0.003717	0.003057
FK 3 465 658	1.464500	65.77	0.007063	1.466186	65.57	0.007110	0.001258	0.002176	0.003597	0.003767	0.003102
FK 5 487 704	1.487490	70.41	0.006924	1.489143	70.22	0.006966	0.001246	0.002144	0.003513	0.003662	0.003006
FK 6 446 674	1.446282	67.43	0.006618	1.447861	67.21	0.006664	0.001174	0.002035	0.003376	0.003540	0.002919
FK 50 486 815	1.486062	81.49	0.005965	1.487486	81.09	0.006012	0.001039	0.001821	0.003057	0.003210	0.002646
FK 51 487 845	1.486561	84.47	0.005760	1.487936	84.07	0.005804	0.001007	0.001762	0.002948	0.003090	0.002542
PK 1 504 669	1.503781	66.92	0.007528	1.505578	66.73	0.007576	0.001349	0.002326	0.003826	0.003998	0.003288
PK 2 518 651	1.518211	65.05	0.007966	1.520112	64.85	0.008020	0.001416	0.002453	0.004059	0.004252	0.003502
PK 3 525 647	1.525419	64.66	0.008126	1.527358	64.46	0.008181	0.001443	0.002501	0.004142	0.004340	0.003578

Density g/cm³	Chemical stability					$\alpha \times 10^7$ -30° to +70 °C	Tg [°C]	Bubble group	τ_i Thickness = 25 mm λ = 400 nm	Special characteristics	Type
	clim. var.	h	V_k	Staining	acid resistance(f)						
2.31	2		3	2	5a	84	390	1	0.989	O	**FK 1** 471 673
2.27	2		4	3	5a	82	362	1	0.96		**FK 3** 465 658
2.45	2		3	1	4	92	464	1	0.991		**FK 5** 487 704
2.28	3		4	2	5b	112	321	1	0.937	p	**FK 6** 446 674
3.63	2			0	5b	143	421	1	0.915	p	**FK 50** 486 815
3.73	2			0	5a	136	405	1	0.92	p	**FK 51** 487 845
2.44	1	1		0	1	62	572	1–2	0.993		**PK 1** 504 669
2.51	3	2		0	2	69	568	0–1	0.992		**PK 2** 518 651
2.59	3	2		0	2	71	567	0–1	0.99		**PK 3** 525 647

Type	n_d	ν_d	n_F-n_C	n_e	ν_e	$n_{F'}-n_{C'}$	n_C-n_r	n_d-n_C	$n_{F'}-n_e$	n_g-n_F	n_h-n_g
PK 50 521 697	1.520539	69.71	0.007467	1.522321	69.50	0.007515	0.001331	0.002303	0.003800	0.003973	0.003268
PSK 2 569 631	1.568730	63.08	0.009016	1.570882	62.85	0.009083	0.001590	0.002765	0.004608	0.004841	0.003999
PSK 3 552 635	1.552322	63.46	0.008704	1.554399	63.23	0.008768	0.001537	0.002671	0.004447	0.004671	0.003859
PSK 50 558 673	1.557529	67.29	0.008286	1.559507	67.03	0.008347	0.001462	0.002542	0.004233	0.004441	0.003664
PSK 51 588 683	1.587841	68.25	0.008613	1.589897	68.04	0.008670	0.001532	0.002654	0.004386	0.004592	0.003782
PSK 52 603 654	1.603100	65.41	0.009220	1.605300	65.14	0.009292	0.001611	0.002817	0.004724	0.004967	0.004105
PSK 53 620 635	1.620140	63.52	0.009763	1.622469	63.23	0.009845	0.001692	0.002972	0.005017	0.005291	0.004382
BK 1 510 635	1.510090	63.46	0.008038	1.512007	63.23	0.008098	0.001416	0.002465	0.004110	0.004319	0.003570

Density g/cm³	Chemical stability					$\alpha \times 10^7$ -30° to +70 °C	Tg [°C]	Bubble group	τ_i Thickness = 25 mm λ = 400 nm	Special characteristics	Type
	clim. var.	h	V_k	Staining	acid resistance(f)						
2.59	2			0	1–2	88	496	1	0.986		PK 50 521 697
3.06	4	3		2	5a	64	608	1	0.974		PSK 2 569 631
2.91	3	2		0	3	62	602	0–1	0.987		PSK 3 552 635
2.93	4		(5)	0	5a	86	543	1	0.85	p	PSK 50 558 673
3.32	4			0	5b	83	597	1–2	0.898	p	PSK 51 588 683
3.38	3			0	5b	85	597	1	0.79	O	PSK 52 603 654
3.60	2			1	5b	94	614	1	0.81	O	PSK 53 620 635
2.46	2	3		0	1	77	547	0–1	0.981		BK 1 510 635

Type	n_d	ν_d	n_F-n_C	n_e	ν_e	$n_{F'}-n_{C'}$	n_C-n_r	n_d-n_C	$n_{F'}-n_e$	n_g-n_F	n_h-n_g
BK 3 498 651	1.498311	65.06	0.007659	1.500139	64.89	0.007707	0.001374	0.002367	0.003892	0.004065	0.003342
BK 6 531 621	1.531130	62.15	0.008546	1.533168	61.92	0.008611	0.001503	0.002618	0.004373	0.004601	0.003805
BK 7 **517 642**	1.516800	64.17	0.008054	1.518722	63.96	0.008110	0.001429	0.002477	0.004108	0.004309	0.003554
UBK 7 517 643	1.516800	64.29	0.008039	1.518719	64.08	0.008095	0.001427	0.002472	0.004100	0.004302	0.003549
BK 8 520 637	1.520152	63.68	0.008168	1.522101	63.46	0.008227	0.001446	0.002510	0.004169	0.004378	0.003616
BK 10 498 670	1.497821	66.95	0.007436	1.499597	66.77	0.007482	0.001335	0.002299	0.003776	0.003944	0.003242
BK 13 521 628	1.521221	62.77	0.008304	1.523202	62.56	0.008363	0.001471	0.002553	0.004238	0.004451	0.003678
BK 50 510 642	1.510107	64.24	0.007941	1.512002	64.01	0.007999	0.001405	0.002439	0.004055	0.004257	0.003515
BaLK 1 526 600	1.526421	60.03	0.008770	1.528513	59.75	0.008845	0.001520	0.002668	0.004510	0.004768	0.003959

Density g/cm³	Chemical stability: clim. var.	h	V_k	Staining	acid resistance(f)	$\alpha \times 10^7$ -30° to +70 °C	Tg [°C]	Bubble group	τ_i Thickness = 25 mm λ = 400 nm	Special characteristics	Type
2.37	2	1		0	2	53	553	2	0.991	O	**BK 3** 498 651
2.69	3	3		0	2	78	537	0–1	0.99		**BK 6** 531 621
2.51	2	2		0	1	71	559	0	0.991		**BK 7** **517 642**
2.51	2	2		0	1	70	563	0	0.993		**UBK 7** 517 643
2.57	2	2		0	1	74	544	1	0.988		**BK 8** 520 637
2.39	1	1		0	1	58	532	1–2	0.992		**BK 10** 498 670
2.54	2			0	1	68	563	1	0.976		**BK 13** 521 628
2.47	2			0	1	75	555	1–2	0.983		**BK 50** 510 642
2.70	2	3		0	2	91	509	1	0.987		**BaLK 1** 526 600

BaLK
K
ZK BaK
SK
KF BaLF
SSK
LaK
cont'd

Type	n_d	ν_d	n_F-n_C	n_e	ν_e	$n_{F'}-n_{C'}$	n_C-n_r	n_d-n_C	$n_{F'}-n_e$	n_g-n_F	n_h-n_g
BaLK 3 518 603	1.518350	60.28	0.008599	1.520400	60.03	0.008669	0.001500	0.002624	0.004413	0.004657	0.003863
K 3 518 590	1.518230	58.98	0.008787	1.520324	58.71	0.008863	0.001521	0.002672	0.004522	0.004783	0.003977
K 4 519 574	1.518951	57.40	0.009041	1.521106	57.15	0.009118	0.001568	0.002750	0.004651	0.004929	0.004108
K 5 522 595	1.522489	59.48	0.008784	1.524583	59.22	0.008858	0.001524	0.002673	0.004517	0.004777	0.003969
K 7 511 604	1.511121	60.41	0.008461	1.513138	60.16	0.008530	0.001474	0.002580	0.004344	0.004587	0.003807
K 10 501 564	1.501371	56.41	0.008888	1.503488	56.15	0.008967	0.001538	0.002700	0.004580	0.004866	0.004068
K 11 500 614	1.500129	61.44	0.008140	1.502070	61.21	0.008203	0.001428	0.002488	0.004172	0.004405	0.003661
K 50 523 602	1.522572	60.18	0.008683	1.524642	59.94	0.008753	0.001515	0.002650	0.004456	0.004702	0.003899
UK 50 523 604	1.522571	60.38	0.008654	1.524635	60.14	0.008723	0.001512	0.002642	0.004439	0.004683	0.003884

Density g/cm³	Chemical stability					$\alpha \times 10^7$ -30° to +70 °C	Tg [°C]	Bubble group	τ_i Thickness = 25 mm λ = 400 nm	Special characteristics	Type
	clim. var.	h	V_k	Staining	acid resistance(f)						
2.61	2	3		0	1	83	506	1	0.989		BaLK 3 518 603
2.54	1	3		0	1	83	521	1–2	0.981		K 3 518 590
2.63	1	3		0	1	73	507	2–3	0.978	O	K 4 519 574
2.59	1	3		0	1	82	543	0–1	0.984		K 5 522 595
2.53	3	3		0	2	84	513	0–1	0.985		K 7 511 604
2.52	1	2		0	1	65	459	1–2	0.984		K 10 501 564
2.50	1	1		0	1	64	493	2	0.986	O	K 11 500 614
2.62	1	1		0	1	70	553	0	0.988		K 50 523 602
2.62	1	1		0	1	70	554	0	0.991		UK 50 523 604

Type	n_d	ν_d	n_F-n_C	n_e	ν_e	$n_{F'}-n_{C'}$	n_C-n_r	n_d-n_C	$n_{F'}-n_e$	n_g-n_F	n_h-n_g
ZK 1 533 580	1.533149	57.98	0.009196	1.535341	57.71	0.009276	0.001589	0.002793	0.004736	0.005017	0.004176
ZK 5 534 553	1.533750	55.31	0.009651	1.536050	55.02	0.009743	0.001648	0.002915	0.004990	0.005308	0.004433
ZK N7 508 612	1.508469	61.19	0.008310	1.510452	60.98	0.008371	0.001470	0.002552	0.004244	0.004462	0.003690
BaK 1 573 575	1.572500	57.55	0.009948	1.574871	57.27	0.010038	0.001707	0.003014	0.005132	0.005442	0.004528
BaK 2 540 597	1.539960	59.71	0.009043	1.542116	59.44	0.009120	0.001566	0.002750	0.004652	0.004918	0.004082
BaK 4 569 561	1.568830	56.13	0.010135	1.571245	55.85	0.010229	0.001740	0.003069	0.005232	0.005558	0.004639
BaK 5 557 587	1.556710	58.65	0.009492	1.558973	58.37	0.009576	0.001637	0.002882	0.004889	0.005177	0.004303
BaK 6 574 564	1.574440	56.40	0.010186	1.576867	56.12	0.010280	0.001750	0.003085	0.005258	0.005587	0.004667
BaK 50 568 580	1.567737	57.99	0.009790	1.570071	57.75	0.009872	0.001698	0.002980	0.005033	0.005322	0.004421

Density g/cm³	Chemical stability					$\alpha \times 10^7$ -30° to +70 °C	Tg [°C]	Bubble group	τ_i Thickness = 25 mm λ = 400 nm	Special characteristics	Type
	clim. var.	h	V_k	Staining	acid resistance(f)						
2.71	1	1		1	1	75	562	1	0.978		**ZK 1** 533 580
2.76	1	2		0	2	87	534	1 2	0.981		**ZK 5** 534 553
2.49	1	1		0	2	45	528	1	0.963		**ZK N7** 508 612
3.19	2	2		1	4	76	602	1	0.976		**BaK 1** 573 575
2.86	2	2		0	1	80	562	1	0.974		**BaK 2** 540 597
3.10	2	2		0–1	3	70	552	1–2	0.984		**BaK 4** 569 561
3.02	2	2		0	2	78	580	1	0.985		**BaK 5** 557 587
3.10	2	2		0	3	73	578	1	0.976		**BaK 6** 574 564
2.93	1	(2)		0	3	37	629	1	0.944		**BaK 50** 568 580

Type	n_d	ν_d	n_F-n_C	n_e	ν_e	$n_{F'}-n_{C'}$	n_C-n_r	n_d-n_C	$n_{F'}-n_e$	n_g-n_F	n_h-n_g
SK 1 610 567	1.610250	56.71	0.010761	1.612815	56.43	0.010860	0.001844	0.003257	0.005555	0.005896	0.004911
SK 2 607 567	1.607381	56.65	0.010721	1.609936	56.38	0.010819	0.001837	0.003246	0.005534	0.005871	0.004890
SK 3 609 589	1.608810	58.92	0.010332	1.611273	58.66	0.010421	0.001787	0.003141	0.005317	0.005624	0.004672
SK 4 613 586	1.612720	58.63	0.010451	1.615212	58.35	0.010543	0.001806	0.003176	0.005380	0.005694	0.004732
SK 5 589 613	1.589130	61.27	0.009615	1.591424	61.03	0.009691	0.001683	0.002939	0.004927	0.005191	0.004296
SK 6 614 564	1.613750	56.40	0.010882	1.616344	56.11	0.010984	0.001862	0.003292	0.005620	0.005965	0.004968
SK 7 607 595	1.607290	59.46	0.010214	1.609726	59.19	0.010301	0.001770	0.003108	0.005252	0.005551	0.004606
SK 8 611 559	1.611171	55.92	0.010929	1.613776	55.64	0.011031	0.001869	0.003305	0.005646	0.005997	0.005000
SK 9 614 552	1.614050	55.17	0.011130	1.616702	54.88	0.011237	0.001901	0.003362	0.005756	0.006127	0.005118
SK 10 623 569	1.622801	56.90	0.010945	1.625410	56.62	0.011045	0.001879	0.003316	0.005647	0.005991	0.004990

Density g/cm³	Chemical stability					$\alpha \times 10^7$ -30° to +70 °C	Tg [°C]	Bubble group	τ_i Thickness = 25 mm λ = 400 nm	Special characteristics	Type
	clim. var.	h	V_k	Staining	acid resistance(f)						
3.56	2	1		1	5a	61	650	0–1	0.975		**SK 1** 610 567
3.55	2	1		0	3	60	654	1	0.975		**SK 2** 607 567
3.53	3	2		2	5a	63	644	0–1	0.968		**SK 3** 609 589
3.57	3	2		2	5a	64	643	0–1	0.973		**SK 4** 613 586
3.30	3	2		1	4	55	658	0–1	0.98		**SK 5** 589 613
3.60	2	1		2–3	5a	62	648	0–1	0.972		**SK 6** 614 564
3.51	3	2		2	5a	64	636	0–1	0.969		**SK 7** 607 595
3.57	2	1		1–2	5a	60	638	0–1	0.975		**SK 8** 611 559
3.60	2	1		2	5a	60	641	0–1	0.956		**SK 9** 614 552
3.66	3	2		3	5b	70	624	0–1	0.96		**SK 10** 623 569

Type	n_d	ν_d	n_F-n_C	n_e	ν_e	$n_{F'}-n_{C'}$	n_C-n_r	n_d-n_C	$n_{F'}-n_e$	n_g-n_F	n_h-n_g
SK 11 564 608	1.563839	60.80	0.009273	1.566051	60.55	0.009348	0.001614	0.002828	0.004759	0.005018	0.004158
SK 12 583 595	1.583131	59.45	0.009808	1.585470	59.19	0.009892	0.001699	0.002984	0.005044	0.005331	0.004424
SK 13 592 583	1.591809	58.30	0.010151	1.594229	58.02	0.010241	0.001747	0.003079	0.005232	0.005539	0.004604
SK 14 603 606	1.603110	60.60	0.009952	1.605484	60.35	0.010033	0.001734	0.003036	0.005107	0.005387	0.004465
SK 15 623 581	1.622990	58.06	0.010731	1.625548	57.79	0.010825	0.001852	0.003259	0.005527	0.005852	0.004866
SK 16 620 603	1.620411	60.33	0.010284	1.622864	60.08	0.010368	0.001794	0.003138	0.005277	0.005566	0.004613
SK N18 639 554	1.638541	55.42	0.011521	1.641286	55.15	0.011628	0.001975	0.003487	0.005950	0.006328	0.005289
SK 19 613 574	1.613421	57.37	0.010693	1.615970	57.08	0.010791	0.001838	0.003240	0.005516	0.005849	0.004870
SK 20 560 612	1.559630	61.21	0.009143	1.561811	60.96	0.009216	0.001595	0.002790	0.004690	0.004942	0.004092
SK 51 621 603	1.620900	60.31	0.010295	1.623355	60.02	0.010386	0.001775	0.003125	0.005303	0.005614	0.004667

Density g/cm³	Chemical stability					$\alpha \times 10^7$ -30° to +70 °C	Tg [°C]	Bubble group	τ_i Thickness = 25 mm λ = 400 nm	Special characteristics	Type
	clim. var.	h	V_k	Staining	acid resistance(f)						
3.08	2	1		0	2	65	610	0–1	0.974		SK 11 564 608
3.27	2	1		1	4	64	633	0–1	0.975		SK 12 583 595
3.37	2	1		2	5a	68	620	0–1	0.97		SK 13 592 583
3.44	4	2		2–3	5a	60	649	0–1	0.97		SK 14 603 606
3.64	3	2		3	5b	69	634	0–1	0.96		SK 15 623 581
3.58	4	3		4	5b	63	638	0–1	0.97		SK 16 620 603
3.64	3	2		4–5	5b	64	643	0–1	0.90		SK N18 639 554
3.56	2	2		1	5a	64	649	0–1	0.972		SK 19 613 574
3.03	3	2		0	2–3	64	605	1	0.977		SK 20 560 612
3.52	2			3	5a	89	597	0–1	0.77	O	SK 51 621 603

Type	n_d	ν_d	n_F-n_C	n_e	ν_e	$n_{F'}-n_{C'}$	n_C-n_r	n_d-n_C	$n_{F'}-n_e$	n_g-n_F	n_h-n_g
SK 52 639 555	1.638540	55.53	0.011499	1.641281	55.27	0.011603	0.001981	0.003487	0.005929	0.006288	0.005239
KF 1 540 511	1.540411	51.10	0.010575	1.542928	50.81	0.010685	0.001793	0.003178	0.005493	0.005897	0.004981
KF 3 515 547	1.514540	54.70	0.009406	1.516781	54.43	0.009494	0.001613	0.002846	0.004859	0.005172	0.004328
KF 6 517 522	1.517421	52.20	0.009913	1.519780	51.93	0.010010	0.001693	0.002991	0.005135	0.005493	0.004624
KF 9 523 515	1.523412	51.49	0.010166	1.525831	51.22	0.010267	0.001731	0.003062	0.005272	0.005648	0.004762
BaLF 2 571 509	1.570989	50.88	0.011223	1.573660	50.58	0.011342	0.001891	0.003366	0.005837	0.006266	0.005284
BaLF 3 571 529	1.571353	52.93	0.010794	1.573923	52.64	0.010902	0.001829	0.003249	0.005598	0.005981	0.005018
BaLF 4 580 537	1.579569	53.71	0.010790	1.582139	53.43	0.010896	0.001837	0.003253	0.005589	0.005962	0.004995

Density g/cm³	Chemical stability					$\alpha \times 10^7$ -30° to +70 °C	Tg [°C]	Bubble group	τ_i Thickness = 25 mm λ = 400 nm	Special characteristics	Type
	clim. var.	h	V_k	Staining	acid resistance(f)						
3.30	2	2		4	5b	60	624	0–1	0.92		**SK 52** 639 555
2.78	1	3		0	1	88	478	2	0.99		**KF 1** 540 511
2.56	1	4		0	1	81	470	1–2	0.97		**KF 3** 515 547
2.67	1	2		0	1	69	446	1	0.988		**KF 6** 517 522
2.71	1–2	(2)		0	1	68	445	0	0.979		**KF 9** 523 515
3.18	1	2		0	2	82	542	2	0.983		**BaLF 2** 571 509
3.18	1	2		0–1	2–3	81	543	2	0.983		**BaLF 3** 571 529
3.17	2	2		0	3	64	569	1–2	0.977		**BaLF 4** 580 537

Type	n_d	ν_d	n_F-n_C	n_e	ν_e	$n_{F'}-n_{C'}$	n_C-n_r	n_d-n_C	$n_{F'}-n_e$	n_g-n_F	n_h-n_g
BaLF 5 547 536	1.547391	53.63	0.010207	1.549821	53.33	0.010309	0.001735	0.003075	0.005291	0.005652	0.004744
BaLF 6 589 530	1.589041	53.01	0.011111	1.591686	52.73	0.011222	0.001886	0.003347	0.005761	0.006154	0.005164
BaLF 8 554 512	1.553609	51.18	0.010816	1.556183	50.89	0.010930	0.001827	0.003247	0.005623	0.006032	0.005087
BaLF 50 589 514	1.588931	51.37	0.011464	1.591659	51.08	0.011582	0.001941	0.003446	0.005953	0.006376	0.005368
BaLF 51 574 521	1.573932	52.10	0.011017	1.576555	51.81	0.011129	0.001868	0.003314	0.005717	0.006120	0.005149
SSK 1 617 539	1.617201	53.91	0.011448	1.619928	53.62	0.011561	0.001949	0.003453	0.005928	0.006322	0.005295
SSK 2 622 532	1.622301	53.16	0.011707	1.625089	52.87	0.011824	0.001988	0.003526	0.006069	0.006482	0.005438
SSK 3 615 512	1.614836	51.16	0.012018	1.617697	50.86	0.012144	0.002026	0.003606	0.006248	0.006697	0.005641
SSK 4 618 551	1.617650	55.14	0.011201	1.620319	54.86	0.011308	0.001913	0.003384	0.005792	0.006161	0.005145

Density g/cm³	Chemical stability					$\alpha \times 10^7$ -30° to +70 °C	Tg [°C]	Bubble group	τ_i Thickness = 25 mm λ = 400 nm	Special characteristics	Type
	clim. var.	h	V_k	Staining	acid resistance(f)						
2.95	2	2		0	1	81	529	1	0.988		**BaLF 5** 547 536
3.32	1	2		1	5a	67	577	2	0.983		**BaLF 6** 589 530
2.99	1	2		0	1	83	513	2	0.988		**BaLF 8** 554 512
3.11	2			0	1	83	555	1–2	0.975		**BaLF 50** 589 514
3.03	1			0	1	81	527	0–1	0.965		**BaLF 51** 574 521
3.63	2	2		3	5a	63	621	1	0.932		**SSK 1** 617 539
3.67	2	2		2	5a	62	636	0–1	0.977		**SSK 2** 622 532
3.61	2	2		1–2	4	66	615	0–1	0.963		**SSK 3** 615 512
3.63	2	2		1	5a	61	639	0–1	0.972		**SSK 4** 618 551

Type	n_d	ν_d	n_F-n_C	n_e	ν_e	$n_{F'}-n_{C'}$	n_C-n_r	n_d-n_C	$n_{F'}-n_e$	n_g-n_F	n_h-n_g
SSK N5 658 509	1.658440	50.88	0.012940	1.661520	50.59	0.013076	0.002183	0.003883	0.006727	0.007213	0.006085
SSK N8 618 498	1.617720	49.77	0.012412	1.620672	49.48	0.012545	0.002092	0.003721	0.006461	0.006951	0.005888
SSK 50 618 526	1.617954	52.61	0.011745	1.620751	52.32	0.011865	0.001990	0.003533	0.006094	0.006516	0.005475
SSK 51 604 536	1.603608	53.63	0.011254	1.606288	53.35	0.011365	0.001914	0.003392	0.005831	0.006225	0.005223
LaK 3 694 535	1.693500	53.45	0.012975	1.696592	53.17	0.013100	0.002210	0.003916	0.006713	0.007144	0.005965
LaK N6 643 580	1.642499	57.96	0.011085	1.645142	57.70	0.011181	0.001915	0.003368	0.005706	0.006037	0.005015
LaK N7 652 585	1.651601	58.52	0.011134	1.654255	58.27	0.011228	0.001931	0.003390	0.005724	0.006050	0.005023
LaK 8 713 538	1.713003	53.83	0.013245	1.716160	53.61	0.013359	0.002297	0.004028	0.006815	0.007218	0.006008
LaK N9 691 547	1.691002	54.71	0.012631	1.694013	54.48	0.012740	0.002186	0.003839	0.006501	0.006880	0.005722

Density g/cm³	Chemical stability					$\alpha \times 10^7$ -30° to +70 °C	Tg [°C]	Bubble group	τ_i Thickness = 25 mm λ = 400 nm	Special characteristics	Type
	clim. var.	h	V_k	Staining	acid resistance(f)						
3.71	2			3	5b	68	641	0–1	0.84		**SSK N5** 568 509
3.33	2			0–1	3	71	597	0–1	0.915		**SSK N8** 618 498
3.42	3			1	1	74	612	0–1	0.89		**SSK 50** 618 526
3.28	3			0	1	76	616	0–1	0.935		**SSK 51** 604 536
4.35	3		4	4–5	5c	82	610	1–2	0.86	Th (8 %)	**LaK 3** 694 535
3.80	3			4	5c	70	634	1	0.985		**LaK N6** 643 580
3.84	4			2	5c	71	618	1	0.96		**LaK N7** 652 585
3.78	3		5	2	5b	56	640	1–2	0.94		**LaK 8** 713 538
3.55	3			3	5b	63	630	1	0.90		**LaK N9** 691 547

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Type	n_d	ν_d	n_F-n_C	n_e	ν_e	$n_{F'}-n_{C'}$	n_C-n_r	n_d-n_C	$n_{F'}-n_e$	n_g-n_F	n_h-n_g
LaK 10 720 504	1.719999	50.41	0.014282	1.723400	50.17	0.014418	0.002449	0.004318	0.007385	0.007875	0.006603
LaK 11 658 573	1.658302	57.26	0.011497	1.661043	57.01	0.011595	0.001993	0.003498	0.005913	0.006256	0.005196
LaK N12 678 552	1.677900	55.20	0.012280	1.680826	54.93	0.012394	0.002101	0.003715	0.006343	0.006735	0.005610
LaK N13 694 533	1.693501	53.33	0.013004	1.696599	53.05	0.013130	0.002211	0.003922	0.006732	0.007166	0.005985
LaK N14 697 554	1.696800	55.41	0.012575	1.699798	55.19	0.012679	0.002190	0.003832	0.006460	0.006828	0.005671
LaK N16 734 517	1.733500	51.65	0.014202	1.736884	51.42	0.014332	0.002443	0.004303	0.007329	0.007790	0.006504
LaK 17 788 505	1.788470	50.48	0.015620	1.792190	50.24	0.015767	0.002674	0.004723	0.008074	0.008592	0.007181
LaK N18 729 542	1.728750	54.16	0.013455	1.731958	53.94	0.013570	0.002337	0.004095	0.006919	0.007321	0.006087
LaK N19 755 531	1.754960	53.05	0.014231	1.758351	52.82	0.014358	0.002455	0.004318	0.007335	0.007780	0.006479
LaK 21 641 601	1.640496	60.10	0.010658	1.643038	59.85	0.010744	0.001861	0.003255	0.005466	0.005765	0.004778

Density g/cm^3	Chemical stability					$\alpha \times 10^7$ -30° to +70 °C	Tg [°C]	Bubble group	τ_i Thickness = 25 mm λ = 400 nm	Special characteristics	Type
	clim. var.	h	V_k	Staining	acid resistance(f)						
3.81	2		5	2	5b	57	620	1	0.91		**LaK 10** 720 504
3.79	3		5	5	5c	72	616	1–2	0.915		**LaK 11** 658 573
4.10	3			1	5c	76	621	1	0.94		**LaK N12** 678 552
4.24	4			2	5c	84	614	1	0.93		**LaK N13** 694 533
3.63	3			2	5b	55	661	1	0.955	O	**LaK N14** 697 554
4.02	1			2	5b	53	611	1	0.93	O	**LaK N16** 734 517
4.88	2			0	4	59	674	1–2	0.795	Th (23 %) O	**LaK 17** 788 505
4.19	1			0	4	53	623	1–2	0.901	Th (19 %) P	**LaK N18** 729 542
4.46	1			0	4	59	658	1–2	0.894	Th (21 %) O	**LaK N19** 755 531
3.74	4			2	5c	68	627	1	0.965		**LaK 21** 641 601

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Type	n_d	ν_d	n_F-n_C	n_e	ν_e	$n_{F'}-n_{C'}$	n_C-n_r	n_d-n_C	$n_{F'}-n_e$	n_g-n_F	n_h-n_g
LaK 22 651 559	1.651129	55.89	0.011650	1.653905	55.62	0.011757	0.001998	0.003527	0.006014	0.006388	0.005330
LaK 23 669 574	1.668815	57.38	0.011655	1.671594	57.12	0.011757	0.002015	0.003541	0.006000	0.006348	0.005274
LaK 24 697 562	1.696800	56.18	0.012404	1.699758	55.96	0.012504	0.002166	0.003786	0.006364	0.006717	0.005572
LaK 25 691 548	1.691001	54.78	0.012613	1.694007	54.51	0.012731	0.002159	0.003815	0.006515	0.006917	0.005762
LLF 1 548 458	1.548140	45.75	0.011980	1.550987	45.47	0.012118	0.002003	0.003574	0.006261	0.006778	0.005776
LLF 2 541 472	1.540719	47.17	0.011464	1.543444	46.88	0.011591	0.001925	0.003428	0.005980	0.006457	0.005488
LLF 3 560 472	1.560130	47.16	0.011876	1.562953	46.88	0.012008	0.001995	0.003552	0.006195	0.006693	0.005695
LLF 4 561 452	1.561381	45.24	0.012410	1.564329	44.95	0.012554	0.002070	0.003699	0.006491	0.007034	0.006003
LLF 6 532 488	1.531719	48.76	0.010905	1.534312	48.48	0.011022	0.001841	0.003270	0.005676	0.006110	0.005177

Density g/cm³	Chemical stability					$\alpha \times 10^7$ -30° to +70 °C	Tg [°C]	Bubble group	τ_i Thickness = 25 mm λ = 400 nm	Special characteristics	Type
	clim. var.	h	V_k	Staining	acid resistance(f)						
3.86	3			2	5c	72	621	1	0.956		**LaK 22** 651 559
4.00	4			2	5c	79	608	1–2	0.935		**LaK 23** 669 574
3.78	2			0	5b	59	654	1–2	0.89	Th (13 %)	**LaK 24** 697 562
4.32	4			4	5c	79	635	0–1	0.92	Th (10 %)	**LaK 25** 691 548
2.94	1	3		0	1	81	448	0–1	0.99		**LLF 1** 548 458
2.87	1	3		0	1	79	442	1	0.99		**LLF 2** 541 472
2.99	1	2		0	1	70	518	2	0.983		**LLF 3** 560 472
3.02	2	2		0	1	82	465	1–2	0.988		**LLF 4** 561 452
2.81	1	3		0	1	75	422	2	0.989	O	**LLF 6** 532 488